Show all work. Indicate if values determined via shock table or code, if not otherwise obvious from the work. Submit any code used. If an LLM is used, attach a separate chat thread for each problem.

ME 561 students are required to typeset assignments using LaTeX. **ME 461** students receive a 2% bonus for typesetting assignments using LaTeX.

1. (24 pts)
   (a) Newton assumed that the sound-wave process was isothermal rather than isentropic. Determine the size of error made in computing the speed of sound that occurs as a result of this assumption.
   (b) A flash of lightning occurs in the distance. Twenty seconds later, the sound of thunder is heard. The temperature in the area is 23 °C. How far away was the lightning strike? (Use our real understanding of sound waves.)

2. (16 pts) Carbon dioxide flows in a horizontal adiabatic, no-work system. Pressure and temperature at section 1 are 7 atm and 600 K. At a downstream section, $p_2 = 4\,\text{atm}$, $T_2 = 550\,\text{K}$, and the Mach number is $M_2 = 0.90$.
   (a) Compute the velocity at the upstream location.
   (b) What is the entropy change?
   (c) Determine the area ratio A_2/A_1.

3. (16 pts) A supersonic aircraft, flying horizontally at a distance H above the earth, passes overhead. A time interval Δt later, the sound wave from the aircraft is heard. In this time increment, the plane has traveled a distance L. Show that the Mach number of the aircraft can be computed with

$$M = \sqrt{\left(\frac{L}{H}\right)^2 + 1} = \sqrt{\left(\frac{V\Delta t}{H}\right)^2 + 1} \tag{1}$$

   Hint: first show that the Mach angle μ can be expressed as $\tan^{-1}(1/\sqrt{M^2-1})$, and then connect the Mach angle to the geometric parameters H and L.

4. **ME 561 ONLY** (12 pts) Express $\frac{\dot{m}}{A}$ in terms of p_t, T_t, γ, R, and M. Make sure to simplify your expression.

Jared Shahim	Gas Dynamics	Wk 2 HW

Problem #1)

a) Newton Assumed, the sound-wave process was isothermal not isentropic

Determine: the size of the error in computing the speed of sound as a result.

Considering that $a = \sqrt{\gamma g_c RT}$

We know that for an Isothermal process $n = 1 = \gamma \mid c_p = c_v$

& that for an isentropic process $n = \gamma \neq 1 \mid c_p \neq c_v$

Now considering a sound wave in isentropic process $\gamma = 1.4$
isothermal process $\gamma = 1$

$\therefore$ $a = \sqrt{(1.4)(287)(273)} = 331.2 \text{ m/s} \rightarrow$ isentropic process

$a = \sqrt{(1)(287)(273)} = 280 \text{ m/s} \rightarrow$ isothermal process

$\because$ $\dfrac{280 \text{ m/s}}{331.2 \text{ m/s}} = 0.845$ $\therefore$ an error of ≈ 0.155

or 15.5% error

* the error will vary w/ respect to the fluid mediums (γ) specific weight

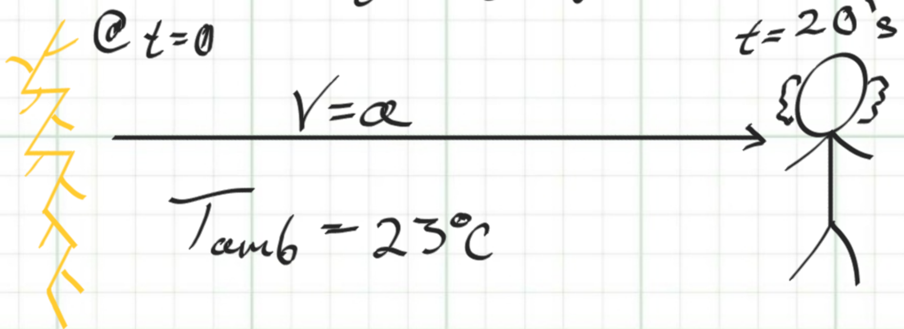
and can be simplified to $\dfrac{a_t}{a_s} = \dfrac{\sqrt{\gamma_t}}{\sqrt{\gamma_s}}$

w/ error $= \dfrac{a_t - a_s}{a_s} \times 100\%$ $\quad$ $t \rightarrow$ isothermal

$s \rightarrow$ isentropic.

6) A flash of lightning occurs in the distance. 20's later the sound of thunder is heard. The ambient Temp is 23°C.

Determine: how far away the lightning struck.



__Assuming:__ Perfect Gas $\rightarrow a = \sqrt{\gamma g_c R T}$

$\gamma = 1.4$

$R = 287 \frac{N \cdot m}{kg\, C}$

$g_c = 1 \frac{kg\, m}{N s^2}$

$$\Rightarrow a = \sqrt{(1.4)\left(\frac{\cancel{kg}\,m}{\cancel{N} s^2}\right)\left(287 \frac{\cancel{N} \cdot m}{\cancel{kg}\, °K}\right)(273.15+23)°K}$$

$a = 344.9537 \ m/s \approx 345 \ m/s$

$\therefore d = a\,\Delta t = (344.9537 \ m/s)(20 \ s) = 6899.07 \ m/s \approx$ (6900 m away) or $\approx$ 4.3 miles away

Problem 2

Carbon dioxide → flows in a horizontal adiabatic, no work system.

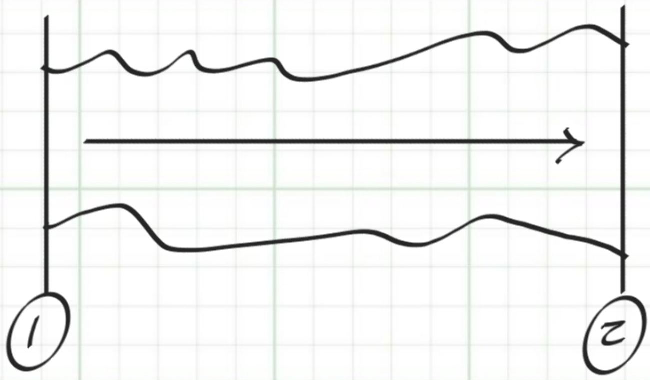
$P_1 = 7\,atm \quad T_1 = 600\,°K$

$P_2 = 4\,atm \quad T_2 = 550\,°K \quad M_2 = 0.90$

a) Compute the velocity @ upstream (i.e at state 1)

b) What is Δs_i

c) Determine the area ratio A_2/A_1



Known: $dQ = dW_s = 0$

$P_1 = 7\,atm \quad T_1 = 600\,°K$

$P_2 = 4\,atm \quad T_2 = 550\,°K \quad M_2 = 0.90$

Analysis:

Equation of state: $p = \rho R T$

Continuity: $\dot{m} = \rho A V = \dot{m}_1 = \dot{m}_2$

$$\Rightarrow \rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \Rightarrow \quad \frac{A_2}{A_1} = \frac{\rho_1 V_1}{\rho_2 V_2}$$

$$\Rightarrow \frac{A_2}{A_1} = \frac{\left(\frac{P_1}{RT_1}\right) V_1}{\left(\frac{P_2}{RT_2}\right) V_2}$$

Since: $M = \frac{V}{a}$ + $a = \sqrt{\gamma g_c R T}$

$$\Rightarrow \frac{A_2}{A_1} = \frac{P_1 T_2 M_1 a_1}{P_2 T_1 M_2 a_2} = \frac{P_1 T_2 M_1 \sqrt{\gamma_1 g_c R T_1}}{P_2 T_1 M_2 \sqrt{\gamma_2 g_c R T_2}} = \frac{P_1 M_1}{P_2 M_2}\left(\frac{\gamma\, T_2}{\gamma\, T_1}\right)^{1/2}$$

Considering the energy Equation

$$\delta q = dw_s + dh_t \Rightarrow h_{t_1} = h_{t_2} \Rightarrow T_{t_1} = T_{t_2}$$

+ Stagnation Pressure Energy Equation

$$\Rightarrow h_t = h\left(1 + \frac{\gamma-1}{2} M_1^2\right)$$

$$\Rightarrow T_t = T\left(1 + \frac{\gamma-1}{2} M_1^2\right)$$

$$\Rightarrow T_1\left(1 + \frac{\gamma-1}{2} M_1^2\right) = T_2\left(1 + \frac{\gamma-1}{2} M_2^2\right)$$

for Ideal Gas γ = constant for CO_2 $\gamma = 1.3$

*Note: for Real Gases γ does change w/ Temp increases.

$$\Rightarrow M_1 = \sqrt{\frac{\left(\frac{T_2}{T_1}\left(1 + \frac{\gamma-1}{2} M_2^2\right) - 1\right)2}{\gamma - 1}}$$

$$M_1 = \sqrt{\frac{\left(\left(\frac{550}{600}\right)\left(1 + (0.15)(0.9)^2\right) - 1\right)2}{0.3}}$$

$$= 0.43234 \approx 0.43$$

$$\therefore V_1 = M_1 a_1 = (0.43234)\sqrt{(1.3)(189)(600)}$$

$$V_1 = 166 \text{ m/s}$$

$$\frac{A_2}{A_1} = \frac{7\text{atm}}{4\text{atm}} \times \frac{0.43}{0.9} \times \left(\frac{550}{600}\right)^{1/2} = 0.805 \approx 0.8$$

Considering That: $ds = \cancel{ds_e}^{0 : dQ=0} + ds_i \Rightarrow ds = ds_i$

$$\therefore \Delta S_{1-2} = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) \Bigg| \begin{matrix} P_{absolute} \\ T_{absolute} \end{matrix}$$

Assume (P and T)'s given are absolute

$$\Delta s_{1-2} = ds_i = 850 \ln\left(\frac{550}{600}\right) - 184 \ln\left(\frac{4}{7}\right)$$

$$ds_i = 31.8 \ J/kg\,K$$

Solutions:

a) $V_1 \approx 166$ m/s $\quad M \approx 0.43$

b) $ds_i = 31.8 \ J/kg\,°K$

c) $\frac{A_2}{A_1} \approx 0.8$

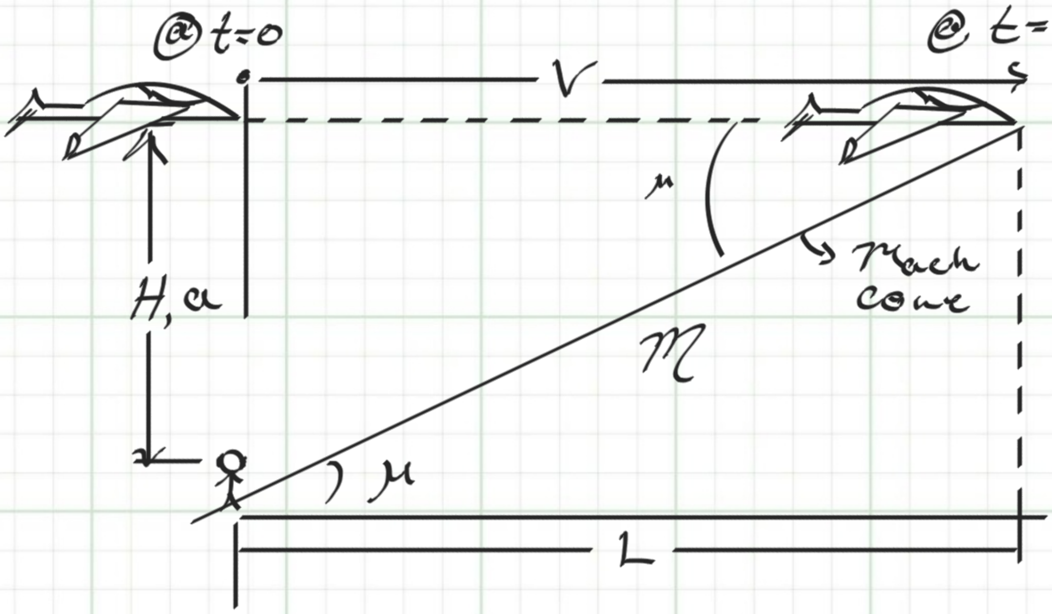
Problem 3

Supersonic aircraft, fly at a distance H above the earth passes overhead. @ Δt later, the sound waves from the aircraft is heard. As well as the plane traveled a distance L.

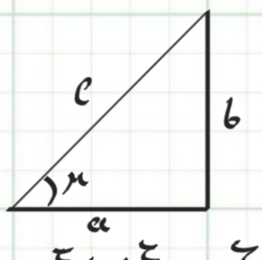
- Show that the Mach # of the Aircraft can be computed w/:

$$M = \sqrt{\left(\frac{L}{H}\right)^2 + 1} = \sqrt{\left(\frac{V\Delta t}{H}\right) + 1}$$

hint: show that the Mach angle μ can be expressed as $\tan^{-1}(1/\sqrt{M^2-1})$, relate μ to H + L



Analysis:



Since L is reached in Δt and H is reached in Δt

$M = 1 = \frac{a}{V} \Rightarrow a = V$ where $\frac{a}{V} \Rightarrow \frac{1}{1}$ ratio

$\Rightarrow L = H$ & $L = V\Delta t$

$a^2 + b^2 = c^2 \Rightarrow a^2 = c^2 - b^2 \quad a = \sqrt{M^2-1}$

$\therefore \tan(\mu) = \frac{b}{a} = \frac{1}{\sqrt{M^2-1}}$

$\Rightarrow \mu = \tan^{-1}\left(\frac{1}{\sqrt{M^2-1}}\right)$

we can also say that

$M^2 = \left(\frac{a}{b}\right)^2 + 1 \Rightarrow M = \sqrt{\left(\frac{L}{H}\right)^2 + 1} = \sqrt{\left(\frac{V\Delta t}{H}\right)^2 + 1}$

Problem 4 * for fun *

- Express: $\frac{\dot{m}}{A}$ in terms of P_t, T_t, γ, R, and M, simplify.

$$\frac{\dot{m}}{A} = \rho V, \quad \rho = \frac{P}{RT}, \quad V = M a \quad \Big| \quad a = \sqrt{\gamma g_c RT}$$

$$\Rightarrow \frac{\dot{m}}{A} = \frac{P M \sqrt{\gamma g_c RT}}{RT} = P M \sqrt{\frac{\gamma g_c}{RT}}$$

∴ Considering the stagnation relations for

$$P_t = P\left(2 + (\gamma - 1) M^2\right)^{\gamma/\gamma-1}$$

&

$$T_t = T\left(2 + (\gamma - 1) M^2\right)$$

$$\therefore \frac{\dot{m}}{A} = \frac{P_t M}{\left(2 + (\gamma-1) M^2\right)^{\gamma/\gamma-1}} \cdot \sqrt{\frac{\gamma g_c \left(2 + (\gamma-1) M^2\right)}{R T_t}}$$

$$\frac{\dot{m}}{A} = P_t M \sqrt{\frac{\gamma g_c \left(2 + (\gamma-1) M^2\right)\left(2 + (\gamma-1) M^2\right)^{\gamma - 1/2\gamma}}{R T_t}}$$

$$\boxed{\frac{\dot{m}}{A} = P_t M \sqrt{\frac{\gamma g_c \left(2 + (\gamma-1) M^2\right)^{3\gamma-1/2\gamma}}{R T_t}}}$$

$$\frac{kg}{m\,s^2}\left(\frac{\frac{kg\,m}{N\,s^2}}{\frac{N\,m}{kg\,K}\,K}\right)^{1/2} = \frac{kg}{m\,s^2}\left(\frac{kg^2}{N^2 s^2}\right)^{1/2}$$

$$= \frac{kg^2}{m\,s^3\,N} = \frac{kg^2}{m\,s^3\,\frac{kg\,m}{s^2}} = \frac{kg}{m^2 s}$$

unit ✓